

Department of Mathematics and Applied Mathematics
Module 3MS: Metric Spaces 2024

Class test 2 solutions

Total Marks: 40

1. State whether each of these statements is true or false. No proof required.
 - (a) The function $f : (\mathbb{R}, d_D) \rightarrow (\mathbb{R}, d_E)$ defined by $f(x) = -x$ is an isometry. FALSE.
 - (b) The function $f : (\mathbb{R}, d_D) \rightarrow (\mathbb{R}, d_E)$ defined by $f(x) = -x$ is continuous. TRUE.
 - (c) The function $g : (\mathbb{N}, d_D) \rightarrow (\mathbb{N}, d_E)$ defined by $g(x) = x$ is a metric isomorphism. FALSE.
 - (d) The function $g : (\mathbb{N}, d_D) \rightarrow (\mathbb{N}, d_E)$ defined by $g(x) = x$ is a topological isomorphism. TRUE.
 - (e) The interval $[1, 95]$ is a complete subset of $(\mathbb{R}, d_E)$. FALSE.
 - (f) The interval $[1, 95]$ is a complete subset of $(\mathbb{R}, d_D)$. TRUE.
 - (g) The empty set $\emptyset$ is a compact subset of (any) metric space (X, d) . TRUE.
[1x7=7]

2.
 - (a) Suppose that $x_n \rightarrow x$ in a metric space (X, d) . Let $\epsilon > 0$ be given.
There exists $N \in \mathbb{N}$ such that, for any $n \in \mathbb{N}$, $n \geq N \implies d(x_n, x) < \frac{\epsilon}{2}$.
Then, for any $n, m \in \mathbb{N}$,
 $m \geq n \geq N \implies d(x_n, x_m) \leq d(x_n, x) + d(x, x_m) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.
 - (b) True.
Suppose that A is a closed subset of a complete metric space (X, d) .
Suppose (a_n) is a Cauchy sequence in A .
Then (a_n) is a Cauchy sequence in X .
Since X is complete, $a_n \rightarrow x$ for some $x \in X$.
Since A is closed, $x \in A$, as required.
 - (c) False.
Counter-example: $(\mathbb{R}, d_E)$ is a complete space; $A = \mathbb{N}$ is a closed subset of $\mathbb{R}$ and $(x_n) = (n)$ is a sequence in A which does not have a convergent subsequence and so A is not compact.

[2+2+3=7]

3. Let X be the set of positive integers. You are given, and do not need to check, that ρ is a metric on X , where

$$\rho(k, \ell) = \left| \frac{1}{k^2} - \frac{1}{\ell^2} \right|.$$

- (a) Let $x_n = n$. We show that (x_n) is Cauchy in (X, ρ) .

$$\text{For } n, m \in \mathbb{N}, \text{ we have } \rho(x_n, x_m) = \rho(n, m) = \left| \frac{1}{n^2} - \frac{1}{m^2} \right|.$$

Let $\epsilon > 0$ be given. Choose $N \in \mathbb{N}$ such that $\frac{1}{n^2} < \epsilon$.

Then $m \geq n \geq N$ implies that $\rho(x_n, x_m) < \epsilon$.

- (b) No, (X, ρ) is not complete.

The sequence (x_n) above is a Cauchy sequence that does not converge.

Proof of this: For $n, k \in X$, $\rho(n, k) = \left| \frac{1}{n^2} - \frac{1}{k^2} \right| \rightarrow \frac{1}{k^2}$ as $n \rightarrow \infty$. Since this limit is not zero, the sequence (x_n) does not converge to any member of X .

[3+3=6]

4. (a) $0 < k < 1$

- (b)

$$\begin{aligned} d(x_{n+1}, x_n) &= d(f(x_n), f(x_{n-1})) \\ &\leq k d(x_n, x_{n-1}) \\ &= k d(f(x_{n-1}), f(x_{n-2})) \\ &\leq k^2 d(x_{n-1}, x_{n-2}) \\ &\vdots \\ &\leq k^{n-1} d(x_2, x_1). \end{aligned}$$

- (c) Since $0 < k < 1$, $k^{n-1} \rightarrow 0$ as $n \rightarrow \infty$. So $\frac{k^{n-1}}{1-k} d(x_2, x_1) \rightarrow 0$ also.

So, for all $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \implies \frac{k^{n-1}}{1-k} d(x_2, x_1) < \epsilon$.

But then $m \geq n \geq N \implies d(x_n, x_m) < \epsilon$.

So (x_n) is Cauchy.

- (d) The space (X, d) was assumed to be complete, so every Cauchy sequence in it converges.

- (e) Since f is a contraction, it is continuous, and thus preserves convergent sequences.

- (f) The expression “limits are unique” means that if $x_n \rightarrow a$ and $x_n \rightarrow b$ in a metric space (X, d) , then $a = b$.

Proof: Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$.

For any $n \in \mathbb{N}$, $0 \leq d(a, b) \leq d(a, x_n) + d(x_n, b)$.

But $d(a, x_n) \rightarrow 0$ and $d(x_n, b) \rightarrow 0$, so $d(a, x_n) + d(x_n, b) \rightarrow 0$.

Taking limits as $n \rightarrow \infty$ gives $d(a, b) = 0$. [1+3+2+1+1+3=11]

5. (a) Suppose $x_n \rightarrow x$ with respect to ρ . Then $\rho(x_n, x) \rightarrow 0$.

Now $0 \leq d(x_n, x) \leq \rho(x_n, x)$ for all n .

Taking limits as $n \rightarrow \infty$ gives $d(x_n, x) \rightarrow 0$, and so $x_n \rightarrow x$ with respect to d .

Suppose $x_n \rightarrow x$ with respect to d . Then $d(x_n, x) \rightarrow 0$.

Now $0 \leq \rho(x_n, x) \leq 4d(x_n, x)$ for all n .

Taking limits as $n \rightarrow \infty$ gives $\rho(x_n, x) \rightarrow 0$, and so $x_n \rightarrow x$ with respect to ρ .

- (b) Possible answers:

(X, d) and (X, ρ) have the same open sets.

(X, d) and (X, ρ) have the same closed sets.

For any $A \subseteq X$, the closure of A in (X, d) is the same as the closure of A in (X, ρ) .

For any $A \subseteq X$, the interior of A in (X, d) is the same as the interior of A in (X, ρ) .

[3+4=7]

6. Suppose X has Property P and $A \subseteq X$.

Let $\epsilon > 0$ be given.

By assumption, there exists a finite $F \subseteq X$ such that $\bigcup_{t \in F} B_X(t; \frac{\epsilon}{2}) = X$.

Let $K = \{x \in F \mid B_X(x; \frac{\epsilon}{2}) \cap A \neq \emptyset\}$.

Note that since K is a subset of F , K is finite.

For each $x \in K$, choose $a_x \in B_X(x; \frac{\epsilon}{2}) \cap A$.

Note that $d(a_x, x) < \frac{\epsilon}{2}$ for all $x \in K$.

We show that $\bigcup_{x \in K} B_A(a_x; \epsilon) = A$:

Take $a \in A$.

Then $a \in B_X(y; \frac{\epsilon}{2})$ for some $y \in F$.

So $a \in B_X(y; \frac{\epsilon}{2}) \cap A$.

So $y \in K$.

Now $d(a, a_y) \leq d(a, y) + d(y, a_y) < \frac{\epsilon}{2} + \frac{\epsilon}{2}$.

So $a \in B_A(a_y; \epsilon)$ and $y \in K$, as desired.

[2]